

Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

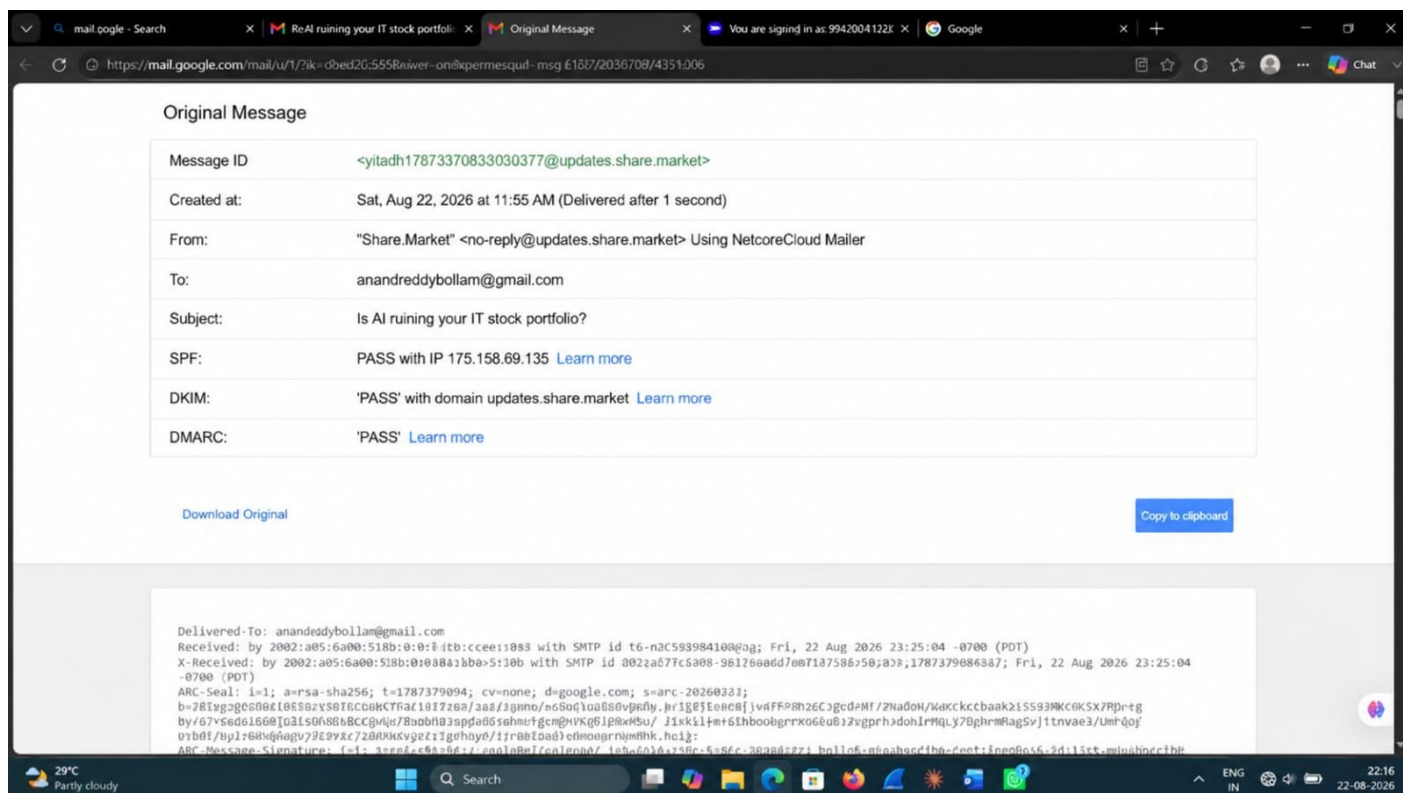
DATE:

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."



2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.

- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - The sending server's hostname/IP address.
 - The receiving server's hostname/IP address.
 - The date and time the email passed through.

Email Header Analyzer

Paste Header:

```
Delivered-To: anandreddybollam@gmail.com
Received: by 2002:a05:6200:5108:20b0:658:cee:5fb9 with SMTP id t8-n2csp1884160qna; Fri, 21 Aug 2026 23:25:04 -0700 (PDT)
X-Received: by 2002:a05:6200:588:8d:8d:5b:5b with SMTP id d2e1472fca58a854f9aa2dfbm116738175b3a.5.1787379904587; Fri, 21 Aug 2026 23:25:04 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1787379904; cv=none; d=google.com; s=arc-20260327; b=DijLYHgf9DBFEol89vC59mKo3VCPKaLltZva/Z2P33iBok/5+ehCiwUlmSmyU6O
47+glTDsvtBdJ+KfBDh22CJgod7M77NaddM/xacKcCbqkA3Js593MKcCOsXJRp+tg OV+0rYY41uLoaRjU1GJ0hdkTBCCQAQlqmqmUSm35p0pX8+6MM+hg+HgnOCQlJpMw5bw/
ik+Nyh+m+KshbD9e9poxGec6U2VgmhJAtd/MqY17PhtmgMVg8yTtnVqa9E/UmTqog yEkmTJAq7o0d6kMqnyvaRzYU/CZdt0RWygZZ7TgphRyo/7IPohbzDBF2HsRaVtTmdDFNhK Rslg==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe-post:feedback-id:list-unsubscribe: content-transfer-encoding:mime-version:message-id:subject:from:to
.date:dkim-signature:dkim-signature; bh=CKYeyTmX2hnZp10w/6Bp/epwMqHavucoDrB2iY58U=: h=HNBNBWeXuLINcRECTOCTH:5J6GTN9pXQlBwYHYD2338=:
b=eCs3Z0HdQ8nRp9qOCmu/ikJdHcJF18wFQ4+2GIUVYnUhnfBsGR5BEI99F5s5F7pEHZx 6ye80iFwrEzRsMttWscWj0tEDYAJheiu4OABpNYIh9xSEs9nGo2KLfUN328NvMEaVB
XT3bjGuWlaxVHZwti0MnpM8bulldYLYQYFMz+ldAnkVAnkXkgfUyjat+R3/DvxL66u44 I5kwU0y7ZTdX2mOohGDBUludYyFsYmrcCfilybwpSrNMAeaOPJM0a0P4xwJPFSLKfgh
```

Analyze Header

ABOUT EMAIL HEADERS

This tool will make email headers human readable by parsing them according to RFC 822. Email headers are present on every email you receive via the Internet and can provide valuable diagnostic information like hop delays, anti-spam results and more. If you need help getting copies of your email headers, [just read this tutorial](#).

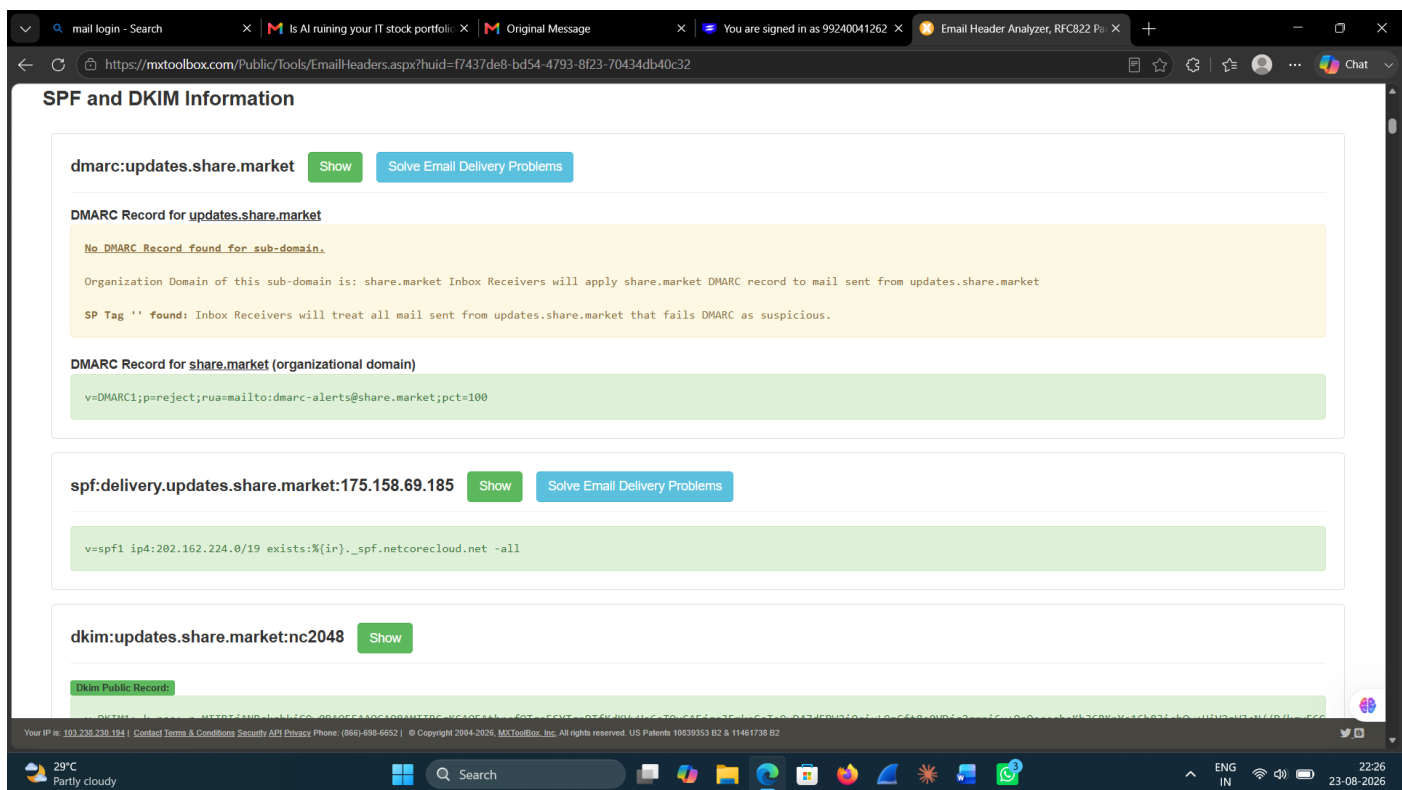
5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.

- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.



7. Analyze the ‘Message-ID’

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.
-

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.

mail login - Search | Is AI ruining your IT stack portfolio? | Original Message | You are signed in as 99240041262 | Email Header Analyzer, RFC822 P... | +

https://mxtoolbox.com/Public/Tools/EmailHeaders.aspx?huid=f7437de8-bd54-4793-8f23-70434db40c32

Header Name	Header Value
Delivered-To	anandreddybolam@gmail.com
X-Received	by 2002.a05.6400:b88-b0:845.bda6.574b with SMTP id d2e1a72fca58-85f19aa2d0bmf673881750d3a.5.1787379904587; Fri, 21 Aug 2026 23:25:04 -0700 (PDT)
ARC-Seal	b1: amsa-sha256, t=1787379904, c=none, d=google.com, s=arc-20260327, b=DjLjyP9G8FvEi0eC8ymhu3JVCRAJiZvAZ2933Bok6-rehCVuAuhSmYU69.4T7gU[2500b5-4h6uho2oQyufTj77Nea5SkaecKcC8eKJ5593MkqCC8uwrp+g-Ov+OY7Y41jLdeRqJ330hdk1BCCAGAA8romUSMa2oP8B-M44b=9mze=3+13524-3pmRi-rlbWp+16R0u66Bnd0KORh5C93pgm8GAM86uY7790H1T4bY0D7TPVQRE9V1Tops.R081C68WIPAHlqRdZbct1uZ50pC0u3uZ6Z7TqHqRq=O8PCC08a35079Y0w8f7N0H0K Raep;
ARC-Message-Signature	v=1; amsa-sha256, c=relaxedrelaxed; d=google.com; s=arc-20260327; h=let-subscribe-post-back id:let-subscribe, content-transfer-encoding:signed-message, id:etbdcfronto, oipk:etm-signature; dkim-signature: b=CiKyeTmK2h7p210w6BpRfepwMplahuc0r0zZY593k; b=cl2Yh0DnLXds0Bop8j8.6B86E5P0u7EYfP3RCR6nbnBoml09ODUJOCU0YfYAAhKv-lwQ0Y950H6B0KfngJ3h9B228gYJumbdxQWhe22pmBh13Qwlf6a0JYn1ngm=VGNdhbz304eGMCavY350zpcKtHhe0htL280tAluicG193XQD_0vZht1mfc2ZuxhGZX+5EGAX=ChdCtT0218h4<1XPWYX06JH0UC5BY9JBRAGIS0bb0RPHB0uBom=866j0uAH92PIM1J0UruDm0RkeYRY.dZ2RkYQAOVQqJkpp5w3150rYHude=
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Authentication-Results	mx.google.com; dkim-pass header l=@updates.share.market header s=rc2048 header b=cl2Yh0DnLXds0Bop8j8.6B86E5P0u7EYfP3RCR6nbnBoml09ODUJOCU0YfYAAhKv-lwQ0Y950H6B0KfngJ3h9B228gYJumbdxQWhe22pmBh13Qwlf6a0JYn1ngm=VGNdhbz304eGMCavY350zpcKtHhe0htL280tAluicG193XQD_0vZht1mfc2ZuxhGZX+5EGAX=ChdCtT0218h4<1XPWYX06JH0UC5BY9JBRAGIS0bb0RPHB0uBom=866j0uAH92PIM1J0UruDm0RkeYRY.dZ2RkYQAOVQqJkpp5w3150rYHude=
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To	<anandreddybolam@gmail.com>
From	Share Market <no-reply@updates.share.market>
Subject	Is AI ruining your IT stack portfolio?
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Content-Type	multipart/alternative; boundary="-----_1787379830187815665"
Content-Transfer-Encoding	binary
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X-Fncd	181468-17873730838360377-0
X-Intme	1787379830
X-Traffic-Type	181468-2
X-Mailer	Netcore Cloud Mailer
List-Unsubscribe-Post	List-Unsubscribe=One-Click

Your IP is 203.39.230.180 | Contact Us | Contact Us | Phone: (800) 900-4847 | © Copyright 2004-2026, MXToolbox, Inc. All rights reserved. US Patent 7,030,032 & 7,180,714 E2

29°C Partly cloudy | Search | ENG IN | 22:27 23-08-2026

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</html>
-----_1787379830187815665-----
```

Permanently forget this email header

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

Successfully Completed. The email header was extracted, parsed, and verified. SPF, DKIM, and DMARC checks confirmed message authenticity without spoofing.